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DISPENSING ASSEMBLY FOR A WASHING MACHINE APPLIANCE

Abstract

A washing machine appliance includes a wash tub positioned within a cabinet and defining a wash chamber, a wash basket rotatably mounted within the wash tub for receiving a load of clothes, and a dispensing assembly mounted within the cabinet for selectively adding wash fluid into the wash tub. The dispensing assembly includes a dispenser housing fluidly coupled to the wash tub through a discharge port, a shower plate positioned above the dispenser housing and defining a plurality of water supply apertures and one or more reservoirs, the one or more reservoirs comprising a main wash reservoir, a detergent drawer slidably mounted below the shower plate, the detergent drawer defining one or more detergent chambers, and a first cold water valve, a second cold water valve, and a hot water valve configured to supply either cold water or hot water to the main wash reservoir.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to washing machine appliances, or more specifically, to dispensing assemblies of washing machine appliances.

BACKGROUND OF THE INVENTION

[0002] Washing machine appliances generally include a tub for containing water or wash fluid, e.g., water and detergent, bleach, and/or other wash additives. A basket is rotatably mounted within the tub and defines a wash chamber for receipt of articles for washing. During normal operation of such washing machine appliances, the wash fluid is directed into the tub and onto articles within the wash chamber of the basket. The basket or an agitation element can rotate at various speeds to agitate articles within the wash chamber, to wring wash fluid from articles within the wash chamber, etc. During a spin or drain cycle, a drain pump assembly may operate to discharge water from within sump.

[0003] Certain conventional washing machine appliances include dispensing assemblies that are configured to dispense water, wash fluid, and/or other additives into the wash chamber at various stages of the operating cycle to facilitate the cleaning of articles located therein. These dispensing assemblies may include a detergent drawer that includes multiple reservoirs for holding different additives and a water supply that selectively supplies water into the one or more reservoirs to flush the additives into the wash tub at desired times in the wash cycle.

[0004] However, conventional dispensing assemblies utilize a large number of components to regulate the fluid flow, e.g., such as multiple water valves, hoses, clamps, etc. For example, conventional dispensing assemblies use a separate cold water valve for each detergent reservoir, plus at least one additional hot water valve. Other dispensing assemblies rely on a water flow diverter to direct incoming water into different channels within the dispensing assembly. These components are costly, increase assembly time and complexity, and provide additional failure points where leaks or malfunctions may occur.

[0005] Accordingly, a washing machine appliance having a dispensing assembly with a minimal number of components would be desirable. More specifically, a dispensing assembly that reliably supplies wash additives with minimal cost and complexity would be particularly beneficial.

BRIEF DESCRIPTION OF THE INVENTION

[0006] Aspects and advantages of the invention will be set forth in part in the following description, or may be apparent from the description, or may be learned through practice of the invention.

[0007] In one exemplary embodiment, a washing machine appliance is provided, including a wash tub positioned within a cabinet and defining a wash chamber, a wash basket rotatably mounted within the wash tub for receiving a load of clothes, and a dispensing assembly mounted within the cabinet for selectively adding wash fluid into the wash tub. The dispensing assembly includes a dispenser housing fluidly coupled to the wash tub through a discharge port, a shower plate positioned above the dispenser housing and defining a plurality of water supply apertures and one or more reservoirs, the one or more reservoirs comprising a main wash reservoir, a detergent drawer slidably mounted below the shower plate, the detergent drawer defining one or more detergent chambers, and a first cold water valve, a second cold water valve, and a hot water valve configured to supply either cold water or hot water to the main wash reservoir.

[0008] In another exemplary embodiment, a dispensing assembly for providing a flow of wash fluid into a wash tub of a washing machine appliance is provided. The dispensing assembly includes a dispenser housing fluidly coupled to the wash tub through a discharge port, a shower plate positioned above the dispenser housing and defining a plurality of water supply apertures and

one or more reservoirs, the one or more reservoirs comprising a main wash reservoir, a detergent drawer slidably mounted below the shower plate, the detergent drawer defining one or more detergent chambers, and a first cold water valve, a second cold water valve, and a hot water valve configured to supply either cold water or hot water to the main wash reservoir.

[0009] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0011] FIG. 1 provides a perspective view of a washing machine appliance according to an example embodiment of the present subject matter.

[0012] FIG. 2 provides a front view of the example washing machine appliance of FIG. 1 with a door in the open position according to an example embodiment of the present subject matter.

[0013] FIG. 3 provides a side cross-sectional view of the example washing machine appliance of FIG. 1 according to an example embodiment of the present subject matter.

[0014] FIG. 4 provides a rear perspective view of a dispensing assembly of the example washing machine appliance of FIG. 1 according to an example embodiment of the present subject matter.

[0015] FIG. 5 provides a perspective view of a dispenser housing of the example dispensing assembly of FIG. 4 according to an example embodiment of the present subject matter.

[0016] FIG. 6 provides a perspective view of a detergent drawer of the example dispensing assembly of FIG. 4 according to an example embodiment of the present subject matter.

[0017] FIG. 7 provides another perspective view of the example detergent drawer of FIG. 6 according to an example embodiment of the present subject matter.

[0018] FIG. 8 provides a top view of a shower plate of the example dispensing assembly of FIG. 4 providing cold water to a pre-wash reservoir according to an example embodiment of the present subject matter.

[0019] FIG. 9 provides a top view of a shower plate of the example dispensing assembly of FIG. 4 providing cold water to a main wash reservoir according to an example embodiment of the present subject matter.

[0020] FIG. 10 provides a top view of a shower plate of the example dispensing assembly of FIG. 4 providing hot water to a main wash reservoir according to an example embodiment of the present subject matter.

[0021] FIG. 11 provides a top view of a shower plate of the example dispensing assembly of FIG. 4 providing warm water to a bleach reservoir according to an example embodiment of the present subject matter.

[0022] FIG. 12 provides a top view of a shower plate of the example dispensing assembly of FIG. 4 providing cold water to a fabric softener reservoir according to an example embodiment of the present subject matter.

[0023] FIG. 13 provides a top view of a shower plate of the example dispensing assembly of FIG. 4 providing warm water to a main wash reservoir according to an example embodiment of the present subject matter.

[0024] Repeat use of reference characters in the present specification and drawings is intended to

represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0025] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0026] As used herein, the terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. For example, the approximating language may refer to being within a 10 percent margin.

[0027] Referring now to the figures, FIG. 1 is a perspective view of an exemplary horizontal axis washing machine appliance **100**, FIG. 2 is a front view of washing machine appliance **100**, and FIG. 3 is a side cross-sectional view of washing machine appliance **100**. As illustrated, washing machine appliance **100** generally defines a vertical direction V, a lateral direction L, and a transverse direction T, each of which is mutually perpendicular, such that an orthogonal coordinate system is generally defined. Washing machine appliance **100** includes a cabinet **102** that extends between a top **104** and a bottom **106** along the vertical direction V, between a left side **108** and a right side **110** along the lateral direction, and between a front **112** and a rear **114** along the transverse direction T.

[0028] Referring to FIGS. 2 and 3, a wash basket **120** is rotatably mounted within cabinet **102** such that it is rotatable about an axis of rotation A. A motor **122**, e.g., such as a pancake motor, is in mechanical communication with wash basket **120** to selectively rotate wash basket **120** (e.g., during an agitation or a rinse cycle of washing machine appliance **100**). Wash basket **120** is received within a wash tub **124** and defines a wash chamber **126** that is configured for receipt of articles for washing. The wash tub **124** holds wash and rinse fluids for agitation in wash basket **120** within wash tub **124**. As used herein, “wash fluid” may refer to water, detergent, fabric softener, bleach, or any other suitable wash additive or combination thereof. Indeed, for simplicity of discussion, these terms may all be used interchangeably herein without limiting the present subject matter to any particular “wash fluid.”

[0029] Wash basket **120** may define one or more agitator features that extend into wash chamber **126** to assist in agitation and cleaning articles disposed within wash chamber **126** during operation of washing machine appliance **100**. For example, as illustrated in FIG. 3, a plurality of ribs **128** extends from basket **120** into wash chamber **126**. In this manner, for example, ribs **128** may lift articles disposed in wash basket **120** during rotation of wash basket **120**.

[0030] Referring generally to FIGS. 1 through 3, cabinet **102** also includes a front panel **130** which defines a chamber opening **132** that permits user access to wash basket **120** of wash tub **124**. More specifically, washing machine appliance **100** includes a door **134** that is positioned over chamber opening **132** and is rotatably mounted to front panel **130**. In this manner, door **134** permits selective access to chamber opening **132** by being movable between an open position (FIG. 2) facilitating access to a wash tub **124** and a closed position (FIG. 1) prohibiting access to wash tub **124**.

[0031] A window **136** in door **134** permits viewing of wash basket **120** when door **134** is in the

closed position, e.g., during operation of washing machine appliance **100**. Door **134** also includes a handle (not labeled) that, e.g., a user may pull when opening and closing door **134**. Further, although door **134** is illustrated as mounted to front panel **130**, it should be appreciated that door **134** may be mounted to another side of cabinet **102** or any other suitable support according to alternative embodiments.

[0032] Referring again to FIG. 3, wash basket **120** also defines a plurality of perforations **140** in order to facilitate fluid communication between an interior of basket **120** and wash tub **124**. A sump **142** is defined by wash tub **124** at a bottom of wash tub **124** along the vertical direction V. Thus, sump **142** is configured for receipt of and generally collects wash fluid during operation of washing machine appliance **100**. For example, during operation of washing machine appliance **100**, wash fluid may be urged by gravity from basket **120** to sump **142** through plurality of perforations **140**.

[0033] A drain pump assembly **144** is located beneath wash tub **124** and is in fluid communication with sump **142** for periodically discharging soiled wash fluid from washing machine appliance **100**. Drain pump assembly **144** may generally include a drain pump **146** which is in fluid communication with sump **142** and with an external drain **148** through a drain hose **150**. During a drain cycle, drain pump **146** urges a flow of wash fluid from sump **142**, through drain hose **150**, and to external drain **148**. More specifically, drain pump **146** includes a motor (not shown) which is energized during a drain cycle such that drain pump **146** draws wash fluid from sump **142** and urges it through drain hose **150** to external drain **148**.

[0034] A spout (such as supply conduit **218**) may be configured for directing a flow of fluid into wash tub **124**. For example, supply conduit **218** may be in fluid communication with a water supply **154** (FIG. 2) in order to direct fluid (e.g., clean water or wash fluid) into wash tub **124**. Supply conduit **218** may also be in fluid communication with the sump **142**. For example, pump assembly **144** may direct wash fluid disposed in sump **142** to supply conduit **218** in order to circulate wash fluid in wash tub **124**.

[0035] As illustrated in FIG. 3, a detergent drawer **156** is slidably mounted within front panel **130**. Detergent drawer **156** receives a wash additive (e.g., detergent, fabric softener, bleach, or any other suitable liquid or powder) and directs the fluid additive to wash tub **124** during operation of washing machine appliance **100**. According to the illustrated embodiment, detergent drawer **156** may also be fluidly coupled to supply conduit **218** to facilitate the complete and accurate dispensing of wash additive.

[0036] In addition, a water supply valve **158** may provide a flow of water from a water supply source (such as a municipal water supply **154**) into detergent dispenser **156** and into wash tub **124**. In this manner, water supply valve **158** may generally be operable to supply water into detergent dispenser **156** to generate a wash fluid, e.g., for use in a wash cycle, or a flow of fresh water, e.g., for a rinse cycle. It should be appreciated that water supply valve **158** may be positioned at any other suitable location within cabinet **102**. In addition, although water supply valve **158** is described herein as regulating the flow of “wash fluid,” it should be appreciated that this term includes, water, detergent, other additives, or some mixture thereof.

[0037] A control panel **160** including a plurality of input selectors **162** is coupled to front panel **130**. Control panel **160** and input selectors **162** collectively form a user interface input for operator selection of machine cycles and features. For example, in one embodiment, a display **164** indicates selected features, a countdown timer, and/or other items of interest to machine users.

[0038] Operation of washing machine appliance **100** is controlled by a controller or processing device **166** (FIG. 1) that is operatively coupled to control panel **160** for user manipulation to select washing machine cycles and features. In response to user manipulation of control panel **160**, controller **166** operates the various components of washing machine appliance **100** to execute selected machine cycles and features.

[0039] Controller **166** may include a memory and microprocessor, such as a general or special purpose microprocessor operable to execute programming instructions or micro-control code

associated with a cleaning cycle. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. In one embodiment, the processor executes programming instructions stored in memory. The memory may be a separate component from the processor or may be included onboard within the processor. Alternatively, controller **166** may be constructed without using a microprocessor, e.g., using a combination of discrete analog and/or digital logic circuitry (such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like) to perform control functionality instead of relying upon software. Control panel **160** and other components of washing machine appliance **100** may be in communication with controller **166** via one or more signal lines or shared communication busses.

[0040] During operation of washing machine appliance **100**, laundry items are loaded into wash basket **120** through chamber opening **132**, and washing operation is initiated through operator manipulation of input selectors **162**. Wash tub **124** is filled with water, detergent, and/or other fluid additives, e.g., via supply conduit **218** and or detergent drawer **156**. One or more valves (e.g., water supply valve **158**) can be controlled by washing machine appliance **100** to provide for filling wash basket **120** to the appropriate level for the amount of articles being washed and/or rinsed. By way of example for a wash mode, once wash basket **120** is properly filled with fluid, the contents of wash basket **120** can be agitated (e.g., with ribs **128**) for washing of laundry items in wash basket **120**.

[0041] After the agitation phase of the wash cycle is completed, wash tub **124** can be drained. Laundry articles can then be rinsed by again adding fluid to wash tub **124**, depending on the particulars of the cleaning cycle selected by a user. Ribs **128** may again provide agitation within wash basket **120**. One or more spin cycles may also be used. In particular, a spin cycle may be applied after the wash cycle and/or after the rinse cycle in order to wring wash fluid from the articles being washed. During a final spin cycle, basket **120** is rotated at relatively high speeds and drain pump assembly **144** may discharge wash fluid from sump **142**. After articles disposed in wash basket **120** are cleaned, washed, and/or rinsed, the user can remove the articles from wash basket **120**, e.g., by opening door **134** and reaching into wash basket **120** through chamber opening **132**.

[0042] While described in the context of a specific embodiment of horizontal axis washing machine appliance **100**, using the teachings disclosed herein it will be understood that horizontal axis washing machine appliance **100** is provided by way of example only. Other washing machine appliances having different configurations, different appearances, and/or different features may also be utilized with the present subject matter as well, e.g., vertical axis washing machine appliances.

[0043] Referring still to FIG. **1**, a schematic diagram of an external communication system **170** will be described according to an exemplary embodiment of the present subject matter. In general, external communication system **170** is configured for permitting interaction, data transfer, and other communications between washing machine appliance **100** and one or more external devices. For example, this communication may be used to provide and receive operating parameters, user instructions or notifications, performance characteristics, user preferences, or any other suitable information for improved performance of washing machine appliance **100**. In addition, it should be appreciated that external communication system **170** may be used to transfer data or other information to improve performance of one or more external devices or appliances and/or improve user interaction with such devices.

[0044] For example, external communication system **170** permits controller **166** of washing machine appliance **100** to communicate with a separate device external to washing machine appliance **100**, referred to generally herein as an external device **172**. As described in more detail below, these communications may be facilitated using a wired or wireless connection, such as via a network **174**. In general, external device **172** may be any suitable device separate from washing machine appliance **100** that is configured to provide and/or receive communications, information, data, or commands from a user. In this regard, external device **172** may be, for example, a personal phone, a smartphone, a tablet, a laptop or personal computer, a wearable device, a smart home

system, or another mobile or remote device.

[0045] In addition, a remote server **176** may be in communication with washing machine appliance **100** and/or external device **172** through network **174**. In this regard, for example, remote server **176** may be a cloud-based server **176**, and is thus located at a distant location, such as in a separate state, country, etc. According to an exemplary embodiment, external device **172** may communicate with a remote server **176** over network **174**, such as the Internet, to transmit/receive data or information, provide user inputs, receive user notifications or instructions, interact with or control washing machine appliance **100**, etc. In addition, external device **172** and remote server **176** may communicate with washing machine appliance **100** to communicate similar information.

[0046] In general, communication between washing machine appliance **100**, external device **172**, remote server **176**, and/or other user devices or appliances may be carried using any type of wired or wireless connection and using any suitable type of communication network, non-limiting examples of which are provided below. For example, external device **172** may be in direct or indirect communication with washing machine appliance **100** through any suitable wired or wireless communication connections or interfaces, such as network **174**. For example, network **174** may include one or more of a local area network (LAN), a wide area network (WAN), a personal area network (PAN), the Internet, a cellular network, any other suitable short- or long-range wireless networks, etc. In addition, communications may be transmitted using any suitable communications devices or protocols, such as via Wi-Fi®, Bluetooth®, Zigbee®, wireless radio, laser, infrared, Ethernet type devices and interfaces, etc. In addition, such communication may use a variety of communication protocols (e.g., TCP/IP, HTTP, SMTP, FTP), encodings or formats (e.g., HTML, XML), and/or protection schemes (e.g., VPN, secure HTTP, SSL).

[0047] External communication system **170** is described herein according to an exemplary embodiment of the present subject matter. However, it should be appreciated that the exemplary functions and configurations of external communication system **170** provided herein are used only as examples to facilitate description of aspects of the present subject matter. System configurations may vary, other communication devices may be used to communicate directly or indirectly with one or more associated appliances, other communication protocols and steps may be implemented, etc. These variations and modifications are contemplated as within the scope of the present subject matter.

[0048] Referring now generally to FIGS. **3** through **13**, a fluid circulation assembly **200** that may be used with washing machine appliance **100** will be described according to example embodiments of the present subject matter. In general, fluid circulation assembly **200** may generally be configured for urging a flow of wash fluid (e.g., identified generally by reference numeral **202** in FIG. **3**) throughout washing machine appliance **100**. For example, the flow of wash fluid **202** may be water, detergent, additives, or some mixture thereof. According to example embodiments, fluid circulation assembly **200** may be configured to circulate the flow of wash fluid **202** within washing machine appliance **100** to facilitate cleaning of a load of articles or may be configured to discharge the flow of wash fluid **202** to an external drain **148**. Although fluid circulation assembly **200** will be described below according to an example embodiment, it should be appreciated that variations and modifications may be made while remaining within the scope of the present subject matter.

[0049] According to the illustrated embodiment, fluid circulation assembly **200** may generally include a dispensing assembly **210** that is configured to receive and distribute the flow of wash fluid **202**. For example, according to the illustrated embodiment, dispensing assembly **210** may generally include a dispenser manifold or a dispenser housing **212** that is positioned within or recessed within cabinet **102**, e.g., at a top corner of front panel **130**. According to the illustrated embodiment, dispenser housing **212** may be an open reservoir positioned at a bottom of dispensing assembly **210** and may include angled collecting wall **214** for directing fluids within dispenser housing **212** toward a discharge port **216**. As shown for example in FIGS. **3** through **5**, discharge port **216** may be fluidly coupled to a supply conduit **218** that is fluidly coupled to wash tub **124**. In

this manner, the flow of wash fluid **202** that is passed into dispenser housing **212** may be directed under the force of gravity into wash tub **124**, e.g., to facilitate operation of washing machine appliance **100**.

[0050] As best illustrated in FIGS. **4** and **8** through **13**, dispensing assembly **210** may further include a shower plate **220** and a top cover **222** that is positioned over the shower plate **220** to define a water supply reservoir **224**. Dispensing assembly **210** may further include a plurality of freshwater supply inlets **226** that are positioned at a rear of shower plate **220** for providing flows of hot and/or cold water into water supply reservoir **224**, e.g., from water supply **154**. In general, shower plate **220** may include a plurality of water supply apertures **228** or perforations for discharging the freshwater from water supply reservoir **224** down into dispenser housing **212**. In this manner, fresh water and/or additives may be showered or flooded within dispenser housing **212** where they may be mixed prior to passing into wash tub **124** through supply conduit **218** (e.g., as flow of wash fluid **202**).

[0051] In addition, as described briefly above, dispensing assembly **210** may include a detergent drawer **156** that is slidably mounted within dispenser housing **212** for receiving one or more wash additives or detergents. In this regard, a user may slide detergent drawer **156** out from front **112** of cabinet **102** for supplying wash additives needed for a wash cycle. Detergent drawer **156** may then slide back into dispensing assembly **210** where water supply **154** may selectively dispense fresh water to flush out one or more compartments of detergent drawer **156** and to create the flow of wash fluid **202**.

[0052] As explained briefly above, detergent drawer **156** may be slidably mounted in dispenser housing **212** below shower plate **220**. In addition, detergent drawer **156** may define one or more detergent chambers (e.g., identified herein generally by reference numeral **240**). Detergent chambers **240** may be generally configured for storing one or more wash additives for use during a washing cycle. For example, according to the illustrated embodiment, detergent drawer **156** defines a pre-wash chamber **242**, a bleach chamber **244**, a fabric softener chamber **246**, and a main wash chamber **248**. It should be appreciated that according to alternative embodiments, detergent drawer **156** may define other chambers for receiving other wash additives while remaining within the scope of the present subject matter.

[0053] Shower plate **220** may generally be positioned over detergent drawer **156** when detergent drawer **156** is inserted into dispenser housing **212**. In addition, shower plate **220** may generally divide water supply reservoir **224** into a plurality of reservoirs associated with one or more detergent chambers **240**. In this regard, by flooding or injecting water into one of the plurality of reservoirs defined by shower plate **220**, the water may pass through water supply apertures **228** into a respective detergent chamber **240**. The water may then flush the detergent within detergent chamber **240** into dispenser housing **212** where it may be discharged into wash tub **124** through supply conduit **218** as a flow of wash fluid **202**.

[0054] According to the illustrated embodiment, shower plate **220** may define four reservoirs that correspond to pre-wash chamber **242**, bleach chamber **244**, fabric softener chamber **246**, and main wash chamber **248**. Specifically, shower plate **220** may define a pre-wash reservoir **252**, a bleach reservoir **254**, a fabric softener reservoir **256**, and a main wash reservoir **258**. Accordingly, flooding these reservoirs of shower plate **220** may cause a respective additive to be dispensed into wash tub **124**, e.g., such as a pre-wash agent, bleach, fabric softener, or a main wash detergent. Although an example configuration of chambers **242-248** and reservoirs **252-258** are illustrated and described herein, it should be appreciated that these chambers and reservoirs may be interchangeable or receive alternative additives while remaining within the scope of the present subject matter.

[0055] In general, dispensing assembly **210** may include a plurality of valves (e.g., such as water supply valve **158**) that are fluidly coupled to a water supply (e.g., such as water supply **154**) for providing a flow of water into water supply reservoir **224**. More specifically, according to the illustrated embodiment, dispensing assembly **210** may include a first cold water valve **260**, a

second cold water valve **262**, and a hot water valve **264**. As shown, water valves **260-264** may be fluidly coupled to the inlet ports **226** that are defined on a back wall **268** of shower plate **220**. As explained in more detail below, by selectively operating water valves **260-264**, water may be selectively supplied into water supply reservoir **224** for dispensing each of the wash additives at a desired time of the wash cycle and at a desired fluid temperature.

[0056] Notably, as explained above, conventional dispensing assemblies require a dedicated water valve for each detergent chamber, along with a separate hot water valve. Moreover, conventional dispensing assemblies are unable to provide flexibility in dispensing hot water, cold water, or warm water (e.g., a mixture of cold and hot water) into the respective chambers. By contrast, the presently disclosed shower plate defines multiple merge points **270** carefully designed to merge flows of hot and cold water at desired locations, thereby providing improved versatility of water flow. Moreover, this varied water flow may be achieved with a minimal number of valves (e.g., three valves) and no other complex diverting or flow directing devices.

[0057] For example, referring now briefly to FIG. **8**, second cold water valve **262** may be opened to supply a flow of cold water directly to pre-wash reservoir **252**. As shown in FIG. **9**, opening both first cold water valve **260** and second cold water valve **262** may direct a flow of cold water to main wash reservoir **258** (e.g., by merging two flows of cold water at merge point **270**). As shown in FIG. **10**, opening hot water valve **264** may direct the flow of hot water to main wash reservoir **258**. In addition, as shown in FIG. **13**, opening first cold water valve **260**, second cold water valve **262**, and hot water valve **264** may direct a flow of warm water main wash reservoir **258** (e.g., by merging flows of cold and hot water at different merge points **270**). As shown in FIG. **11**, opening first cold water valve **260** and hot water valve **264** may direct a flow of warm water to bleach reservoir **254**. In addition, as shown FIG. **12**, opening first cold water valve **260** may supply a flow of cold water to fabric softener reservoir **256**. It should be appreciated that other flow paths, merge points, and other fluid supply configurations may be used while remaining within the scope of the present subject matter.

[0058] As explained herein, aspects of the present subject matter are generally directed to a washing machine including a dispensing assembly (e.g., including shower plate assembly, diffuser, dispenser drawer, etc.) in which water streams are merged at the multiple zones for redirecting the path of the water flow. The dispenser may include four regions: a pre-wash reservoir located in the rear left region of the dispenser drawer; a main wash reservoir located in the front left region of the drawer; a bleach reservoir located in the rear right region; and a fabric softener reservoir located in the front right region of the drawer. The dispenser system may utilize only three water valves (e.g., two cold and one hot) to operate. During a pre-washing agent dispensing and the fabric softener dispensing, only one valve may be actuated i.e., the cold water valve “Cold 2” may be actuated to direct water to the pre-wash reservoir, and the cold water valve “Cold 1” may be actuated to direct water to the fabric softener reservoir. During the main wash, water stream from two different water valves (“Cold 1” and “Cold 2”) may be merged to direct the flow of water through the channel leading to the main wash reservoir. Further, the flow of the hot water to the main wash reservoir may be accomplished with the actuation of the hot valve only. During the dispense of the fabric softener, water stream from two different valves may be merged (“Cold 1 and “Hot”) and then the water may flow through the channel leading to the fabric softener reservoir. The dispensing assembly described herein reduces the number of water valves required to dispense a given number of wash system additives. By reducing the number of water valves, the number of hoses and clamps connecting the water valves to the dispenser assembly are also reduced. This results in a lower cost for both purchased materials and for the labor required to assemble the water valves, hoses, and clamps.

[0059] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of

the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A washing machine appliance comprising: a wash tub positioned within a cabinet and defining a wash chamber; a wash basket rotatably mounted within the wash tub for receiving a load of clothes; and a dispensing assembly mounted within the cabinet for selectively adding wash fluid into the wash tub, the dispensing assembly comprising: a dispenser housing fluidly coupled to the wash tub through a discharge port; a shower plate positioned above the dispenser housing and defining a plurality of water supply apertures and one or more reservoirs, the one or more reservoirs comprising a main wash reservoir; a detergent drawer slidably mounted below the shower plate, the detergent drawer defining one or more detergent chambers; and a first cold water valve, a second cold water valve, and a hot water valve configured to supply either cold water or hot water to the main wash reservoir.
2. The washing machine appliance of claim 1, wherein the one or more reservoirs includes four reservoirs.
3. The washing machine appliance of claim 1, wherein inlet ports for the first cold water valve, the second cold water valve, and the hot water valve are defined on a back wall of the shower plate.
4. The washing machine appliance of claim 1, wherein the one or more reservoirs comprise a pre-wash reservoir, a bleach reservoir, a fabric softener reservoir, and the main wash reservoir.
5. The washing machine appliance of claim 4, wherein the second cold water valve is opened to supply a flow of cold water to the pre-wash reservoir.
6. The washing machine appliance of claim 4, wherein the first cold water valve and the second cold water valve are opened to supply a flow of cold water to the main wash reservoir.
7. The washing machine appliance of claim 4, wherein the hot water valve is opened to supply a flow of hot water to the main wash reservoir.
8. The washing machine appliance of claim 4, wherein the first cold water valve, the second cold water valve, and the hot water valve are opened to supply a flow of warm water to the main wash reservoir.
9. The washing machine appliance of claim 4, wherein the first cold water valve and the hot water valve are opened to supply a flow of warm water to the bleach reservoir.
10. The washing machine appliance of claim 4, wherein the first cold water valve is opened to supply a flow of cold water to the fabric softener reservoir.
11. The washing machine appliance of claim 1, wherein the washing machine appliance is a front load washing machine appliance.
12. A dispensing assembly for providing a flow of wash fluid into a wash tub of a washing machine appliance, the dispensing assembly comprising: a dispenser housing fluidly coupled to the wash tub through a discharge port; a shower plate positioned above the dispenser housing and defining a plurality of water supply apertures and one or more reservoirs, the one or more reservoirs comprising a main wash reservoir; a detergent drawer slidably mounted below the shower plate, the detergent drawer defining one or more detergent chambers; and a first cold water valve, a second cold water valve, and a hot water valve configured to supply either cold water or hot water to the main wash reservoir.
13. The dispensing assembly of claim 12, wherein the one or more reservoirs includes four reservoirs.
14. The dispensing assembly of claim 12, wherein the one or more reservoirs comprise a pre-wash

reservoir, a bleach reservoir, a fabric softener reservoir, and the main wash reservoir.

15. The dispensing assembly of claim 14, wherein the second cold water valve is opened to supply a flow of cold water to the pre-wash reservoir.

16. The dispensing assembly of claim 14, wherein the first cold water valve and the second cold water valve are opened to supply a flow of cold water to the main wash reservoir.

17. The dispensing assembly of claim 14, wherein the hot water valve is opened to supply a flow of hot water to the main wash reservoir.

18. The dispensing assembly of claim 14, wherein the first cold water valve, the second cold water valve, and the hot water valve are opened to supply a flow of warm water to the main wash reservoir.

19. The dispensing assembly of claim 14, wherein the first cold water valve and the hot water valve are opened to supply a flow of warm water to the bleach reservoir.

20. The dispensing assembly of claim 14, wherein the first cold water valve is opened to supply a flow of cold water to the fabric softener reservoir.
